

Smart cities will generate vast amounts of data from all connected devices, which will keep growing exponentially. So, it is a fundamental requirement to set up the technical infrastructure in place to store this valuable data. It is equally important to have an abundant and reliable power supply to keep the whole infrastructure working.

Data centres will be the major contributors to the emergence of smart cities, as these can address such crucial requirements as ample connectivity, huge data storage and high compute power for analytics to crunch all data.

The eco-friendly factor

One of the main objectives of smart cities is to address the current environmental challenges. Smart cities promise to improve energy efficiency by bringing in advanced applications that track individual and corporate emissions, and to suggest ways to achieve this. Moreover, organisations need to meet the environment-friendly standards and set up eco-friendly IT infrastructure to comply with the government's Smart Cities initiative.

Modular data centres have an edge in reducing organisational power consumption. Their power usage effectiveness (PUE) figures are expected to be much lower than those of traditional data centres. However, considering the fact that there is always a need for larger, core data centres to deliver organisations' digital services, data centre optimisation processes can be opted for by retuning these with smarter cooling solutions. This can largely increase the energy efficiency and IT capacity of organisations, enabling them to take advantage of the emerging IT landscape.

Shifting to co-located facilities also plays a key role in contributing to the eco-friendly parameters of the smart



cities. Various SMBs and large-sized enterprises that need to expand their infrastructure capacity should look towards hosting co-located data centres, which offer secure and safe access to data. This is also a simple way to improve energy efficiency of organisations.

Data centres contribute to smart cities: an undeniable fact

Data centres bridge the gap between edge and core levels of smart city architecture. Advanced monitoring of data flow and cutting-edge management technologies allow the new data centre environment to be controlled from a single glass pane. Futurists and experts say that data centres are one of the main building blocks that will bring smart cities to reality. **END** 🐧

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OSFY Magazine Attractions During 2019-20

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April 2019	Cybersecurity, Open Source Firewall, Network Security and Monitoring
May 2019	AI, Deep Learning and Machine Learning
June 2019	DevOps Special
July 2019	Blockchain and Open Source
August 2019	Database Management and Optimisation
September 2019	Open Source and IoT
October 2019	Cloud Special: BigData, Hadoop, PaaS, SaaS, IaaS and Cloud
November 2019	Open Source on Windows
December 2019	Open Source Programming (Languages and Tools)
January 2020	Data Security, Storage and Backup
February 2020	Best in the World of Open Source (Tools and Services)